THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

Dec 16, 2016 14:30 pm – 17:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. With a severe shortage in public housing accommodation, the government would like to convert some vacant apartments located in various public housing estates into smaller housing units for young people. To avoid any unnecessary nuisance, it is expected that the works will be carried out over a long period of time.

   a) With reference to the nature of work mentioned above, explain to the government the features of lump sum contract and remeasurement contract and discuss their advantages and disadvantages when applied to this type of alteration work. [17 marks]

   b) The government would also like to seek your advice on the principle and operation of a term contract as well as its suitability for this project. [8 marks]

2. The following table shows the duration and logical relationship of the activities involved in a construction project:

Activity	*Immediately preceding activity*	*Duration*
A	--	4
B	--	3
C	--	2
D	A	2
E	C	5
F	B, D, E	1
G	B, D	6
H	D	5
I	G	7
J	F	7
K	J	6
L	I, K	5
M	K	1
N	K	4
O	I	6
P	H, I (SS2)	9
Q	L	2
R	L	3
S	P	9
T	M, N, O, Q, R	6
U	S	4
V	T	1
W	U, V	2

   a) Based on the principles of the critical path method, prepare a table to show the early and late start and completion dates and work out the total float and free float for each activity. [20 marks]

   b) Describe the measures to be taken by a project manager to monitor and control project progress. [5 marks]

3. a) Advise a hotel group the maximum price they can spend on land acquisition based on the following information.

Gross floor area = 5,000 m^2
Machine rooms, water tank, etc. = 5% of gross floor area
Expected number of rooms = 300
Construction cost = \$18,000/m^2
Construction period = 36 months
Average rate for a guest room = \$1,000/room/day
Anticipated occupancy rate = 75% in the first year
100% afterwards
Operating cost = \$400/room/day
Short-term finance = 10% per annum
Yield = 8% [18 marks]

b) Highlight the major risks associated with the project described in (a) and explain how to avoid or minimise those risks from occurring. [7 marks]

4. a) Explain with relevant example(s) what is qualification, front loading, covered price and tender error, and confirm with valid reasons whether these issues are acceptable in tendering sense. [12 marks]

b) Describe what actions should be taken if the issues mentioned in (a) are discovered in a lump sum contract. [9 marks]

c) Discuss whether the course of actions would be different should front-loading or tender error be discovered in a remeasurement contract. [4 marks]

5. a) A Hong Kong based construction company is keen to participate in projects along the Belt and Road regions. Discuss whether it is suitable to adopt a functional organisation approach when branch offices are first established in the Belt and Road countries for tendering purpose. [8 marks]

b) Explain the characteristics of a project based and matrix organisation and discuss under what circumstance the construction company should consider employing these two organisational settings. [17 marks]

~ END OF PAPER ~